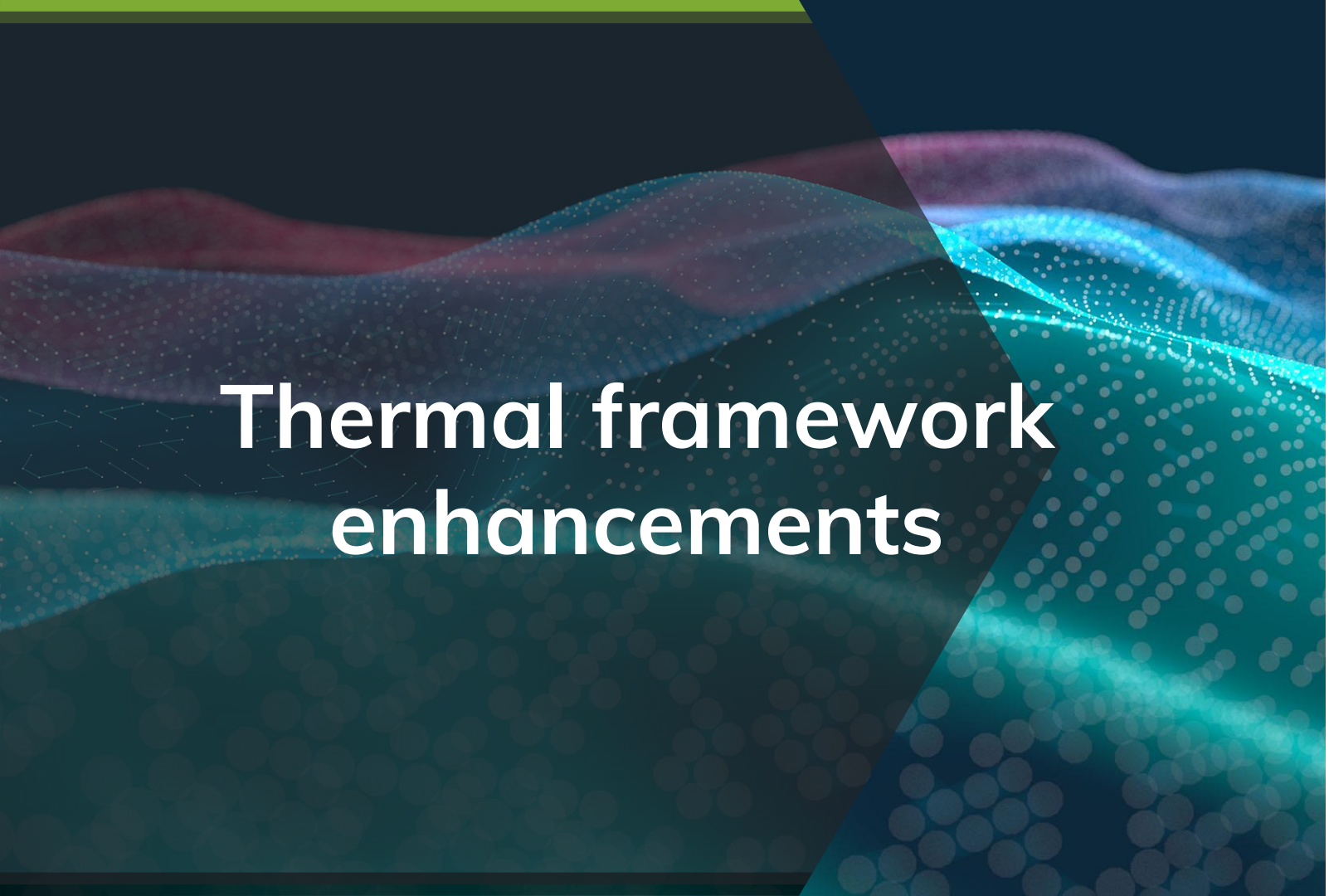
Linux Plumbers Conference 2023 Thermal MC

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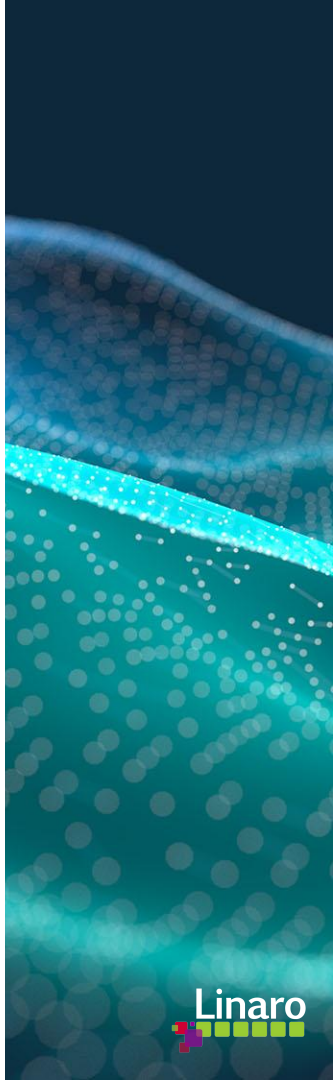


Thermal framework enhancements



Thermal enhancements

- User space dedicated trip points
- Debugfs for the thermal framework
- Improving the step wise governor



New trip point type

Dedicated user space trip point

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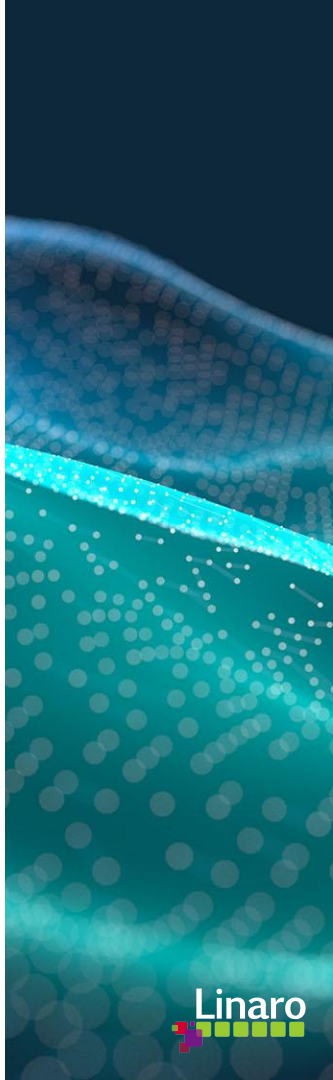
Userspace dedicated trip points

- Most of the drivers have interrupt detection based notifications
- Userspace wants to be notified when a temperature threshold is crossed
- Writable trip points option
  - Passive, active, hot and critical temperature are writable
- Thermal framework is about protecting the silicon, letting userspace playing with the trip points is bad

Userspace dedicated trip points

Proposal : Create a **<user>** trip point type

- The thermal framework can clearly identify it and ignore any action except sending notification to userspace
- Takes benefit of interrupt based synchronous trip violation detection
- Writable trip point Kconfig option can be disabled
- <user> trip points stay writable
- Shall we create multiple <user> trip points?



Thermal debugfs

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Debugfs for the thermal framework

- Very few information to investigate the thermal framework behavior
- Bogus cooling device stats
- We need to know how efficient are the mitigations
  - Depends on ambient temperature
  - Initial temperature
  - Heat capacity headroom
  - Temperature change speed
- Too many overshots = bad for the hardware
- Too many undershots = bad for the performances

Debugfs for the thermal framework

- Directory structure close to sysfs but simplified

```
thermal/
|-- cooling_devices
|   |-- 0
|   |   |-- reset
|   |   |-- time_in_state_ms
|   |   |-- total_trans
|   |   `-- trans_table
`-- thermal_zones
    |-- 0
    |   `-- mitigations
    `-- 1
        `-- mitigations
```


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Mitigation episodes

,-Mitigation at 349988258us, duration=130136ms

trip	type	temp(°mC)	hyst(°mC)	duration	avg(°mC)	min(°mC)	max(°mC)
0	passive	65000	2000	130136	68227	62500	75625
1	passive	75000	2000	104209	74857	71666	77500

,-Mitigation at 272451637us, duration=75000ms

trip	type	temp(°mC)	hyst(°mC)	duration	avg(°mC)	min(°mC)	max(°mC)
0	passive	65000	2000	75000	68561	62500	75000
1	passive	75000	2000	60714	74820	70555	77500

,-Mitigation at 238184119us, duration=27316ms

trip	type	temp(°mC)	hyst(°mC)	duration	avg(°mC)	min(°mC)	max(°mC)
0	passive	65000	2000	27316	73377	62500	75000
1	passive	75000	2000	19468	75284	69444	77500

,-Mitigation at 39863713us, duration=136196ms

trip	type	temp(°mC)	hyst(°mC)	duration	avg(°mC)	min(°mC)	max(°mC)
0	passive	65000	2000	136196	73922	62500	75000
1	passive	75000	2000	91721	74386	69444	78125

Debugfs for the thermal framework

- More information planned

,-Mitigation at **349988258**us, duration=130136ms

trip	type	temp(°mC)	hyst(°mC)	duration	avg(°mC)	min(°mC)	max(°mC)
0	passive	65000	2000	130136	68227	62500	75625
1	passive	75000	2000	104209	74857	71666	77500

```
thermal/
|-- [ … ]
`-- thermal_zones
    `-- 0
        `-- mitigations
            `-- 349988258
                |-- speed
                |-- [ … ]
                `-- stddev
```


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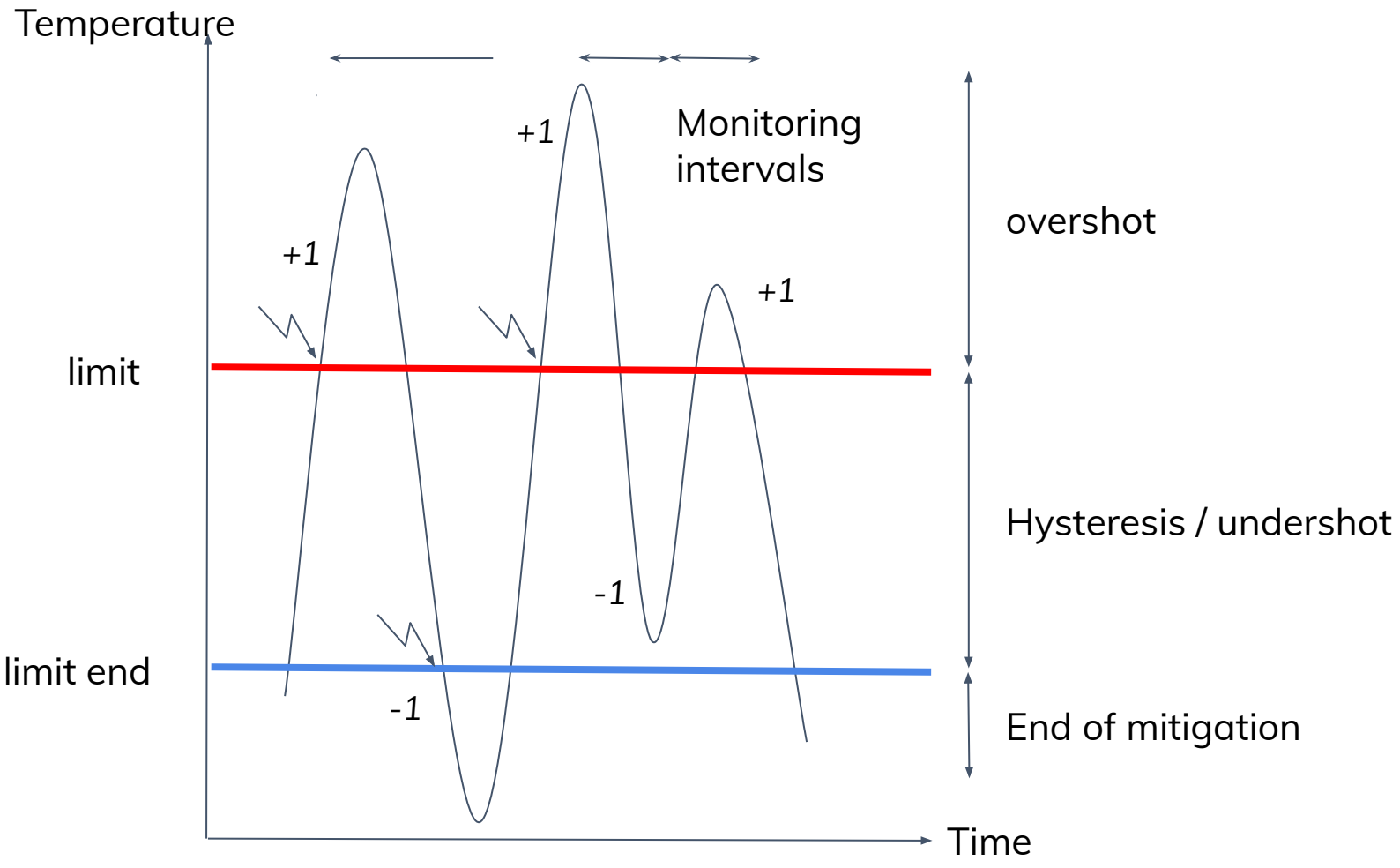
Improving the stepwise governor

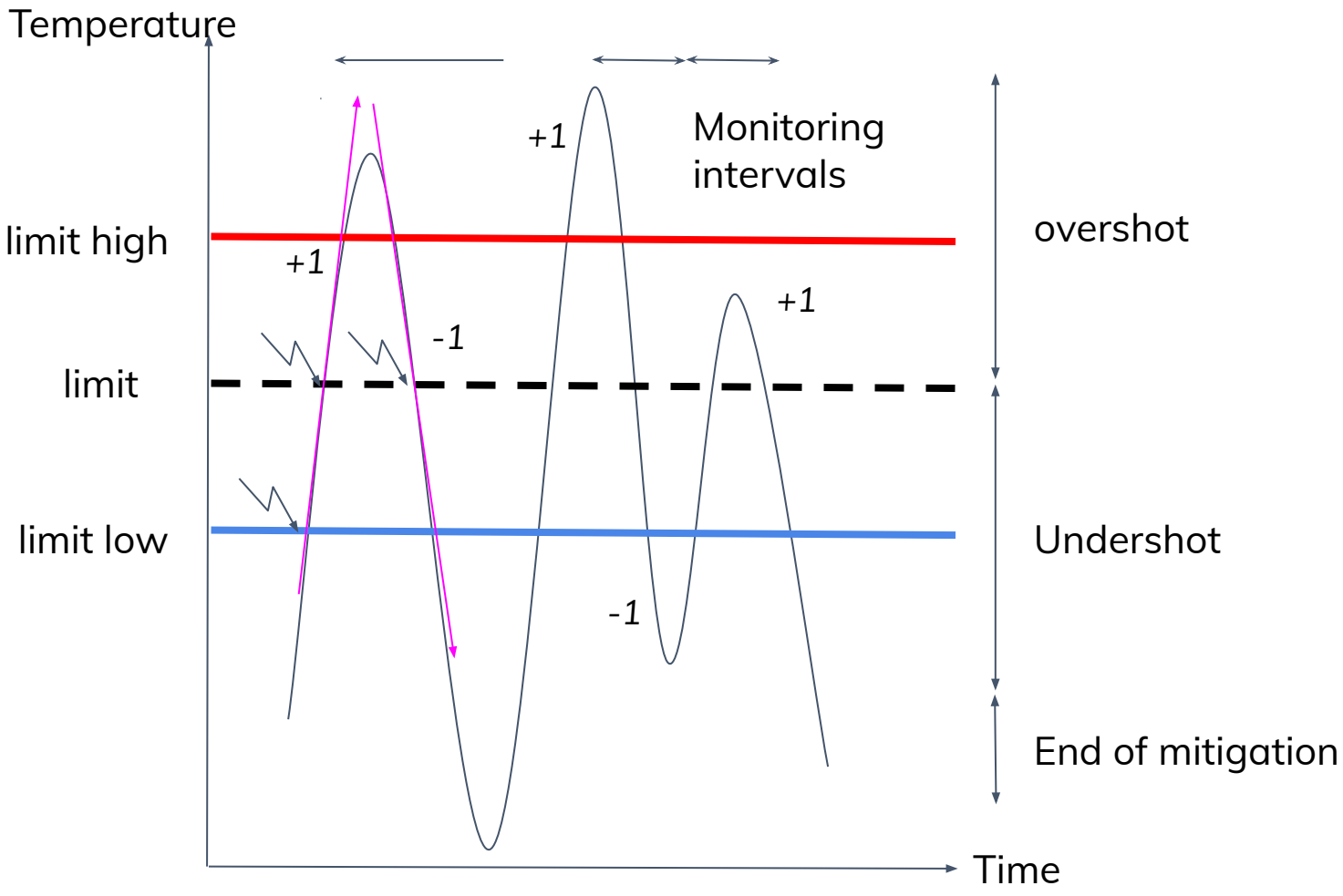
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Stepwise governor improvement

- Problem: stepwise governor does not react fast enough
- Consequences: overshoots and undershoots
- Current solution: Increase sampling

- Proposal:
    - Temperature speed computation
    - Temperature forecast
    - Next trip point crossed anticipated by a timer
    - We keep monitoring anyway


Temperature
Monitoring intervals
+1
+1
+1
limit
limit end
-1
-1
overshot
Hysteresis / undershot
End of mitigation
Time


Temperature
Monitoring intervals
+1
limit high
+1
+1
-1
limit
limit low
-1
overshot
Undershot
End of mitigation
Time

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Thank you

